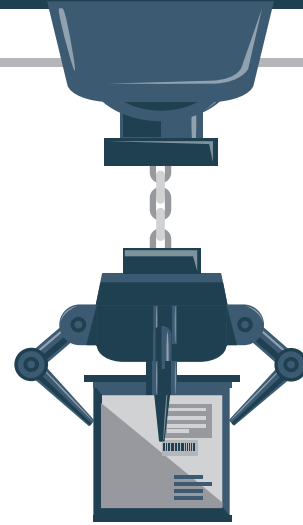




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Multipurpose CNC Machine With Process Optimization

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Department of Electrical Engineering
Third-Year

Project Guide
Prof. Dr. Garima Gurjar

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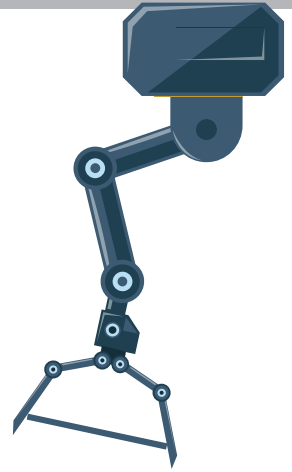
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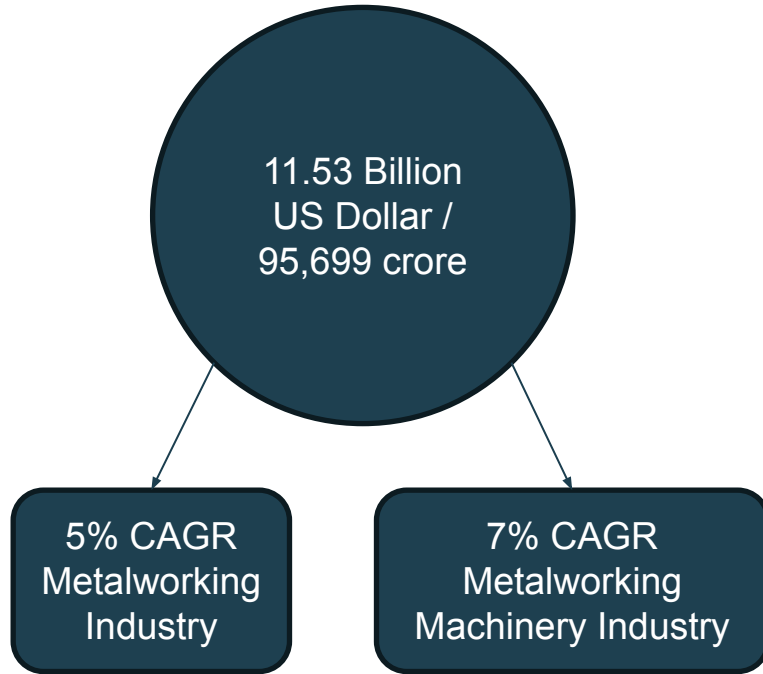
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Conclusion



Current Scenario

Metal Fabrication Market



INDIA

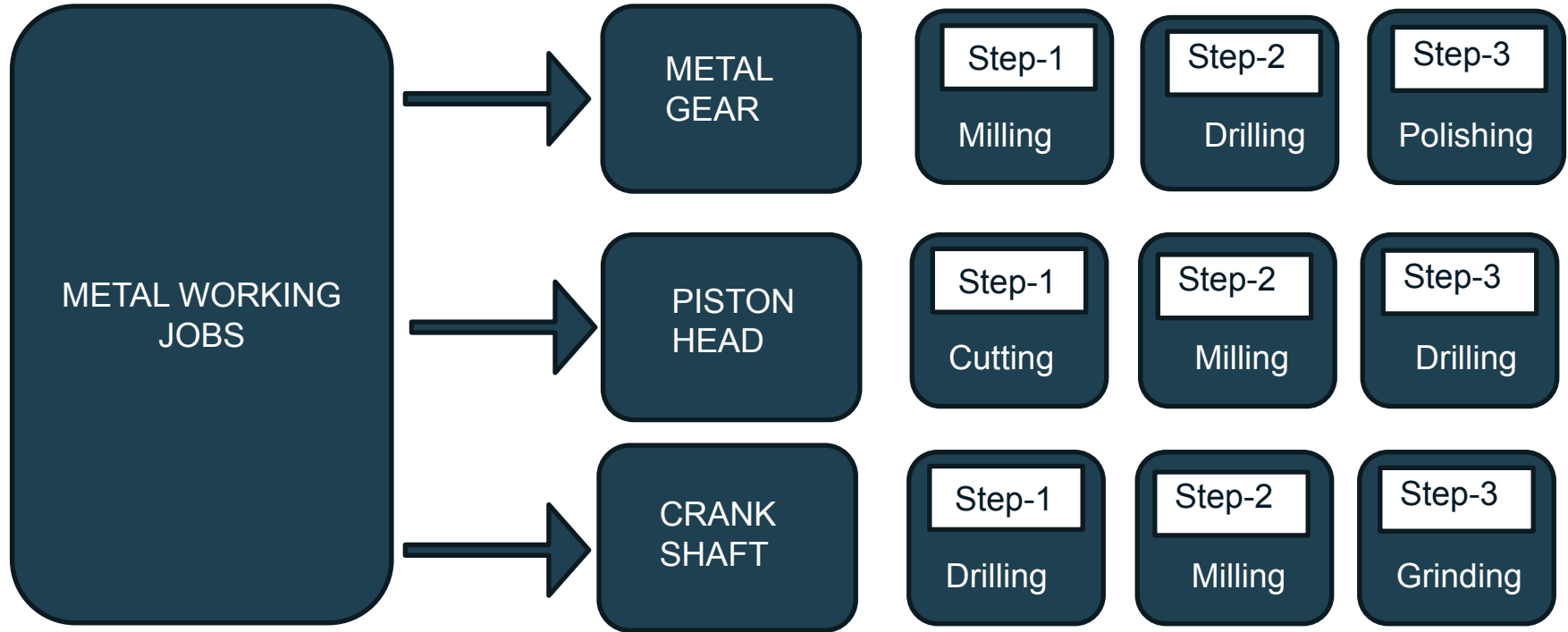
6.4 crore registered MSME

- 30 – 40% registered for metal working and fabrication.
- Daily volume of 2000 ton/ day.
- Utilization of multiple machines.

Isko hatana hai



Problem Statement



HIGHER TIME CONSUMPTION

- Reduced Production
- Delayed process
- Raw material over stacking

REQUIREMENT OF MULTIPLE MACHINES

- Machine set-up delay
- Higher maintenance
- Higher processing time

DOWN TIME OF MACHINE

- Higher down time
- Reduced production time
- Machine start-up losses

HIGHER LABOUR COST

- Increased cost per unit
- Inventory stacking losses
- Lower profit ratio

Market available CNC machine



Introduction to device

Multi – Axes
Motion
Control

Enhanced 3-axis
(X, Y, Z) CNC
platform

Multi – Function
Machine

- Drilling
- Milling
- Grinding
- Cutting
- Engraving

Adaptive
Closed Loop
feedback
system

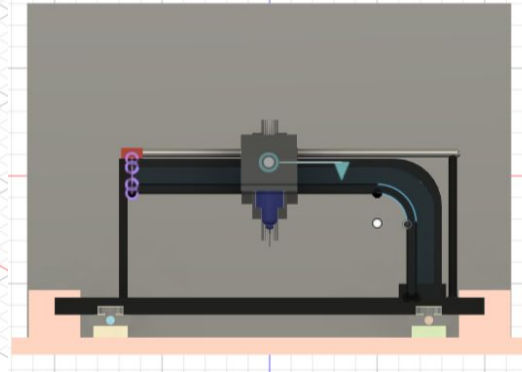
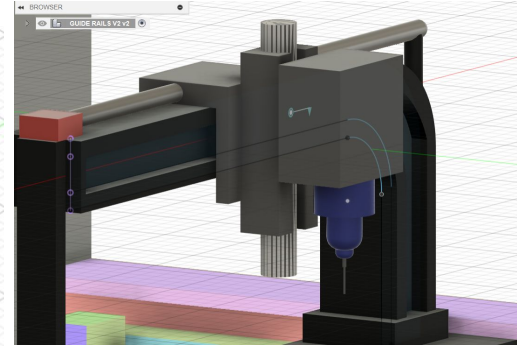
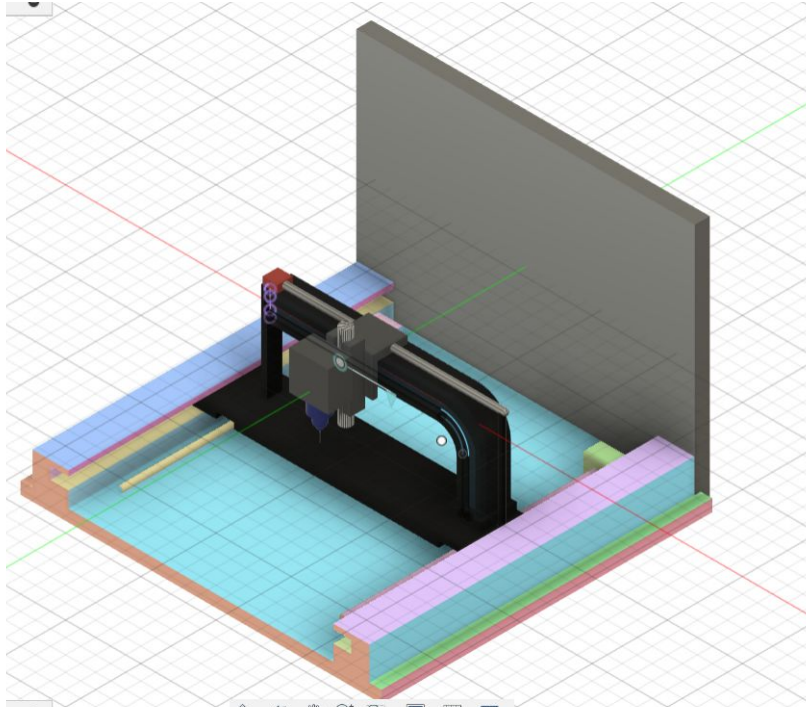
Adaptive feed-rate
control and process
optimization





Proposed Design

(<https://a360.co/4n02v3M>)





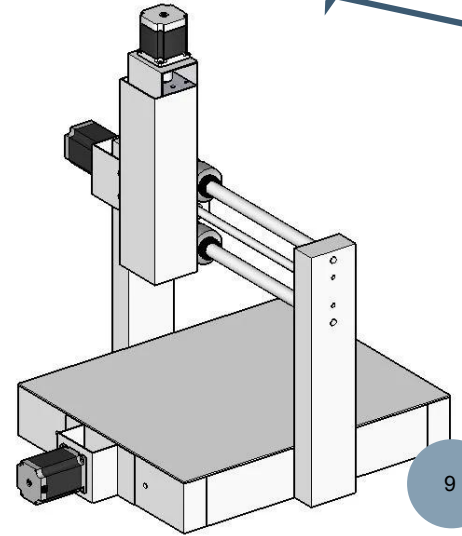
Research & Methodology

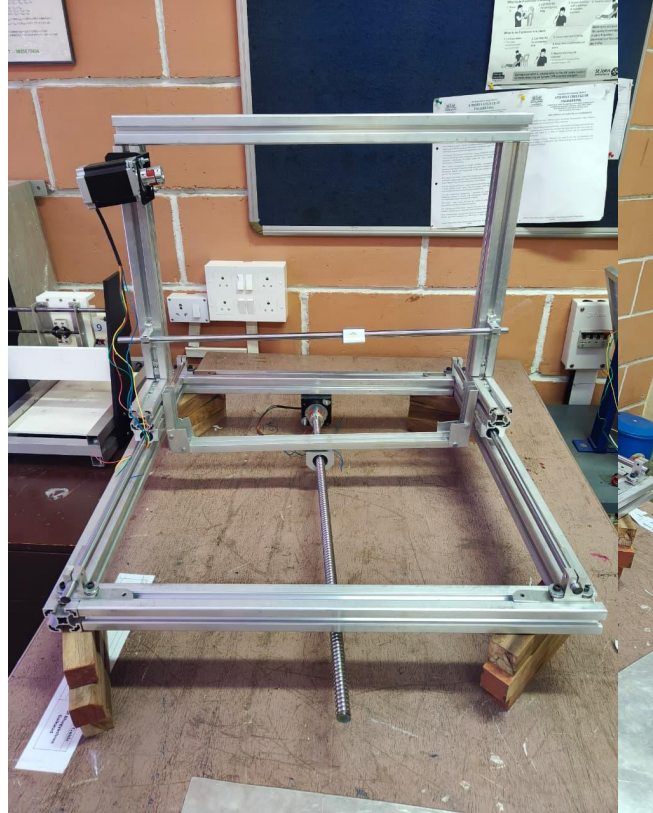
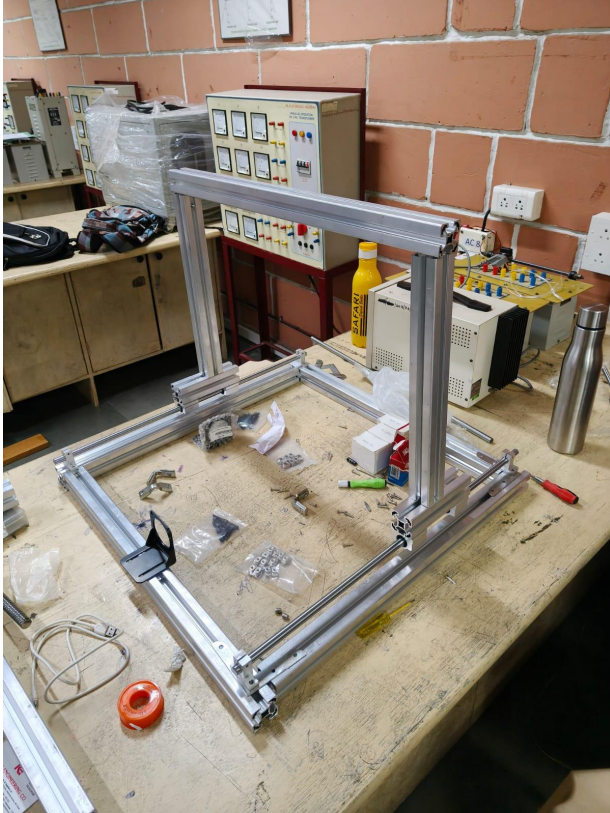
Key Configurations

- Z-axis linear rail + stepper motor
- ESP32-S3 controller + touchscreen
- Spindle load sensor for adaptive control
- Multi-spindle mechanism

Key Challenges solved

- Equipment consolidation
- Space and cost efficiency
- Multi-function mechanism

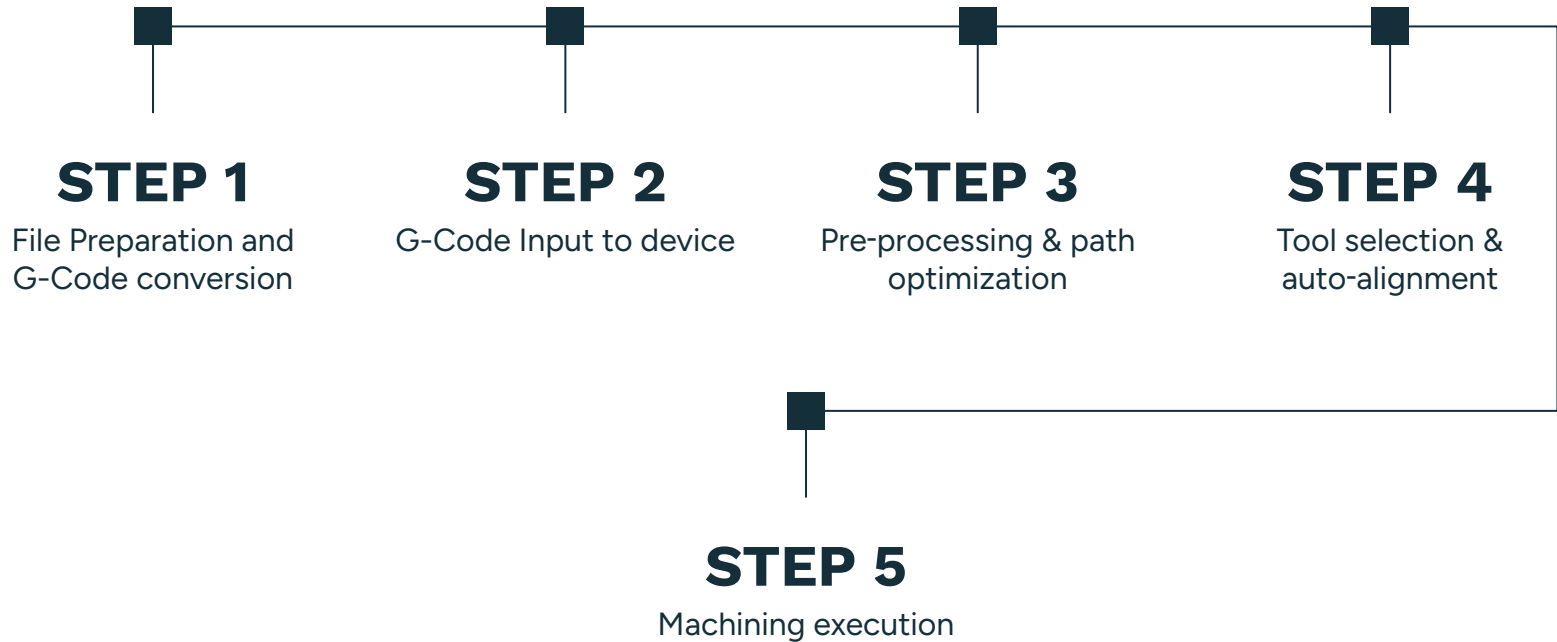




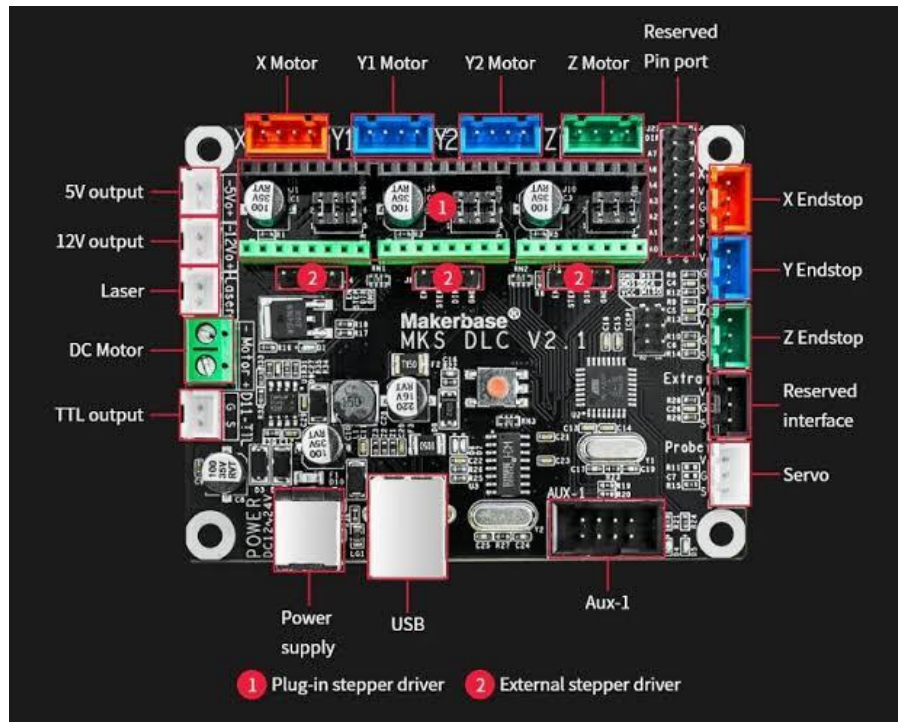
Multipurpose CNC Machine With Process Optimization



Process Flowchart



\$ Controller MKS DLC32





PROS

- 3D Capability
- Compact footprint
- Cost-effective



CONS

- Increased mechanical complexity

Isko hatana hai

Growth Timeline

 Sem 5 – Mechanical Build
(base structure fabrication, assembly)



 Sem 6 – Circuit & Software Design
(electronics integration, control coding)



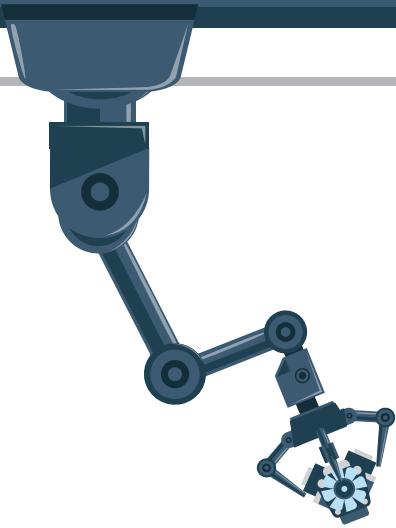
 Sem 7- Advanced Features
(remote monitoring, optimization)



Future Scope

- IoT integration for remote monitoring.
- 3D Printer spindle configuration.
- Automated multi-tool carousel to minimize setup time
- CAD/CAM software plugins for streamlined workflow





Conclusion

- Therefore we are coming with a product which is not only 1/3rd of the market but also reducing the labor & execute the Job less time with multiple features that too in a small-scale fabrication environments.

THANK YOU!

Any
Questions/Feedbacks?